# CONSEQUENCE DISTILLATION FOR TRANSFERABLE AUTOMATIC SPEECH RECOGNITION EVALUATION

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## ABSTRACT

Word Error Rate (WER) remains the standard metric for evaluating Automatic Speech Recognition (ASR) performance, despite treating all recognition errors equally regardless of context. We introduce a consequence-distillation framework that learns from a frozen task evaluator whether errors affect downstream task outputs at two granularities. At the transcript level, Direct Estimate predicts whether a complete ASR hypothesis changes the downstream task output. At the edit level, Edit-Consequence WER (EC-WER) uses counterfactual repairs to predict the consequence of each aligned substitution, deletion, or insertion, providing attribution to individual recognition errors while retaining conventional WER structure. Zero-shot results reveal a granularity tradeoff, offering practitioners a clear choice: Direct Estimate for pure downstream-failure prediction, and EC-WER when interpretable, edit-level error diagnosis is also required. These results demonstrate that consequence distillation can support both failure-oriented and diagnostic ASR evaluation. Code and evaluation scripts are publicly available online at `https://github.com/peridous/EC-WER`.

***Index Terms*—** automatic speech recognition, word error rate, consequence distillation, contextual edit consequence

## 1. INTRODUCTION

Word Error Rate (WER) is the standard metric for evaluating Automatic Speech Recognition (ASR), but it treats every substitution, deletion, and insertion equally [1]. This can be misleading for task-oriented speech systems, where only a subset of recognition errors may affect the intended task output, while others are largely inconsequential. For example, in spoken command understanding, an utterance can be represented by structured labels describing the requested action, object, and location. Changing "bring me some *juice*" to "bring me some *water*" changes the requested object, while changing "bring me *some* juice" to "bring me juice" may leave the intended command unchanged. An evaluation metric should therefore reflect not only how many errors occur, but also how much those errors affect the downstream task.

However, measuring this consequence is challenging because the importance of an error depends on its **context**. Existing approaches often assign importance to individual words, entities, or semantic similarity between complete sentences [2, 3]. However, a word that is generally important may not affect the task in every transcription. Likewise, the effect of correcting one ASR error may depend on the other errors already present in the hypothesis, as illustrated in Fig. 1. Thus, token importance alone does not sufficiently capture the consequence of a recognition error in context. We instead evaluate consequence through downstream tasks.

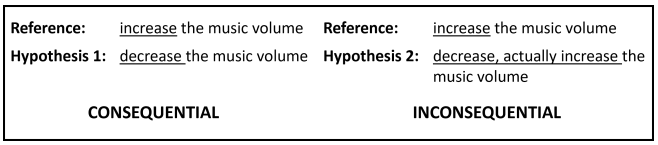



**Fig. 1**: Context dependence of edit consequence. The same focal edit can have different effects depending on the surrounding hypothesis.

Downstream consequence can be modeled at two levels. At the *transcript level*, a model predicts whether the full ASR hypothesis causes a downstream failure. This directly measures task impact, but it does not show which recognition errors caused the failure. At the *edit level*, consequence is predicted for each individual substitution, deletion, or insertion, providing more detailed error attribution.

We study both levels using a consequence-distillation framework. A frozen task evaluator generates training labels but is not needed at test time. For transcript-level distillation, a student predicts whether the full hypothesis changes the downstream task output. For edit-level distillation, we repair one ASR error at a time and train a student to predict whether that repair changes the task output. These edit predictions are then used to form Edit-Consequence WER (EC-WER).

This allows us to compare direct downstream-failure prediction with finer-grained edit-level attribution. We use FSC as the source task for consequence learning because it provides structured semantic outputs. We then test whether this learned consequence transfers beyond FSC through independent FSC targets and zero-shot evaluation on SLURP and VoxPopuli. Our contributions are:

1. We formulate **consequence distillation for ASR evaluation at two levels**: transcript-level prediction of downstream failure and edit-level attribution of individual recognition errors.
2. We introduce **counterfactual edit-consequence distillation** and EC-WER, which predicts the consequence of substitutions, deletions, and insertions in context while preserving WER alignment and normalization.
3. We show that transcript-level distillation is generally preferable when failure prediction is the sole objective, while EC-WER is the better choice whenever edit-level diagnosis is also needed, **with zero-shot transfer across independent tasks and datasets.**

## 2. RELATED WORK

Prior work has extended WER to better reflect recognition-error severity. SASRED uses supervised SimCSE distance to con-

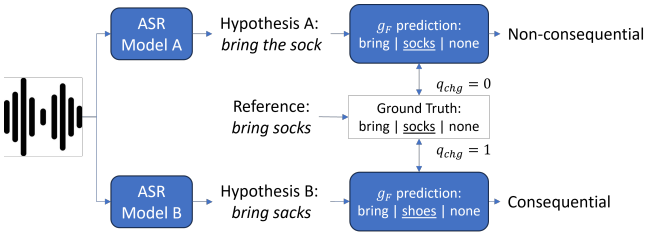



**Fig. 2**: Overview of contextual edit-consequence supervision and EC-WER scoring.

struct weak labels for transcript-level significant-error detection [4]. Entity-focused approaches include Medical WER, which evaluates aligned medical entities [2], and ATENE, which estimates ASR effects on downstream NER [5]. These methods either specialize in particular semantic structures or do not model how an edit's impact changes with transcription context.

A second line of work evaluates semantic similarity between the reference and hypothesis. BERTScore compares contextual token representations [6], while SemDist measures sentence-level representation distance [3]. Aligned Semantic Distance (ASD) instead combines contextual embeddings with dynamic-programming alignment [7]. More recent approaches also use model-based judgments to evaluate transcription meaning [8, 9]. Such metrics provide an overall semantic comparison between the reference and hypothesis, rather than explicitly estimating the consequence of each aligned ASR edit.

Our supervision mechanism is related to erasure and leave-one-out attribution [10]. However, we apply the intervention to each observed ASR error in its specific transcription context. Rather than testing whether a token is important in general, we correct one aligned ASR edit while keeping all other recognition errors unchanged. This produces an edit-level consequence target that allows the same edit to receive different consequence labels depending on the surrounding recognition errors.

## 3. METHOD

We study consequence distillation at two levels. *Transcript-level* distillation directly predicts whether an ASR hypothesis causes downstream failure, while *edit-level* distillation predicts the consequence of each individual recognition error.

### 3.1. Transcript-Level Consequence Distillation

Let $R = (r_1, \ldots, r_N)$ be the normalized reference word sequence of length $N$, and $H = (h_1, \ldots, h_M)$ the corresponding ASR hypothesis of length $M$. Let $g$ denote a frozen task evaluator and $\mathbf{1}[\cdot]$ the indicator function. We define transcript-level downstream failure as

$$z_g(R, H) = \mathbf{1}\left[g(R) \neq g(H)\right]. \tag{1}$$

For transcript-level consequence distillation, we train a student model $P_\phi$, parameterized by $\phi$, that maps the reference–hypothesis pair $(R, H)$ to a probability of downstream failure. Its prediction is

$$\hat{z}_{\text{direct}} = P_\phi\left(z_g = 1 \mid R, H\right). \tag{2}$$

where $\hat{z}_{\text{direct}} \in [0, 1]$ denotes the predicted probability that $H$ changes the downstream output relative to $R$.

We refer to this transcript-level model as *Direct Estimate*. It directly estimates downstream failure but produces a single score for the complete transcript and therefore does not directly provide edit-level attribution.

### 3.2. Contextual Edit Repair

For edit-level distillation, we compute the standard unit-cost minimum-WER alignment between $R$ and $H$, and let $\mathcal{E}(R, H)$ denote the resulting set of substitution, deletion, and insertion errors.

For each error $e \in \mathcal{E}(R, H)$, we construct a counterfactual hypothesis $H^{(-e)}$ by repairing only $e$ while leaving all other recognition errors unchanged. For a substitution, the hypothesis word is replaced by its aligned reference word; for an insertion, the inserted word is removed; and for a deletion, the missing reference word is restored at its aligned position.

We define the edit-consequence label as

$$q_{\text{chg}}(e; R, H) = \mathbf{1}\left[g\left(H^{(-e)}\right) \neq g(H)\right]. \tag{3}$$

Thus, $q_{\text{chg}} = 1$ indicates that repairing the edit changes the downstream prediction, while $q_{\text{chg}} = 0$ indicates that the prediction is preserved.

### 3.3. Contextual Edit-Consequence Prediction

We train a second student model $P_\theta$, parameterized by $\theta$, on the counterfactual labels from Eq. (3). The student receives the complete reference $R$, hypothesis $H$, and aligned edit $e$, and predicts

$$\hat{q}_e = P_\theta\left(q_{\text{chg}} = 1 \mid R, H, e\right). \tag{4}$$

where $\hat{q}_e \in [0, 1]$ denotes the predicted probability that repairing edit $e$ changes the downstream prediction. Because the student conditions on the complete hypothesis, the same focal edit can receive different consequence probabilities in different transcription contexts. After training, the task evaluator $g$ is no longer required for scoring.

### 3.4. Edit-Consequence WER

Conventional WER assigns unit cost to every substitution, deletion, and insertion. EC-WER instead weights each aligned edit by its predicted consequence probability:

$$\text{EC-WER}(R, H) = \frac{1}{N} \sum_{e \in \mathcal{E}(R,H)} \hat{q}_e. \tag{5}$$

EC-WER preserves the standard WER alignment and reference-length normalization while replacing uniform edit costs with learned consequence weights. We use the continuous probability $\hat{q}_e$ rather than a binary decision so that the score retains the student's confidence in each edit.

## 4. EXPERIMENTAL SETUP

### 4.1. Datasets and ASR Models

We evaluate outputs from HuBERT, wav2vec 2.0, WavLM, Whisper-base, and Whisper-tiny [11, 12, 13, 14]. Before alignment, transcripts are lowercased and tokenized with punctuation removed. We evaluate only hypotheses containing at least one recognition error because our goal is to rank the downstream impact of ASR errors rather than distinguish erroneous transcripts from exact matches.

**Table 1**: Comparison of transcript- and edit-level consequence distillation across downstream-failure targets. * indicates a statistically significant difference between Direct Estimate and EC-WER ($p < 0.05$).

| | | AUPRC | | | AUROC | | |
|---|---|---|---|---|---|---|---|
| Data | Target | WER | Direct | EC-WER | WER | Direct | EC-WER |
| FSC | $g_F$ | .6093 | .7996 | **.8245*** | .7111 | .8319 | **.9188*** |
| FSC | $g_B$ | .4985 | **.7949*** | .6699 | .7154 | **.9087*** | .8663 |
| FSC | NLI | .4996 | **.6546*** | .5772 | .7071 | **.8457*** | .7664 |
| SLURP | Full | .9486 | .9640 | **.9644** | .8126 | .8549 | **.8554** |
| SLURP | Intent | .8353 | .8692 | **.8784*** | .7973 | .8302 | **.8344** |
| SLURP | NLI | .8814 | **.9189*** | .9001 | .7891 | **.8517*** | .8213 |
| VoxPop. | NER | .4024 | **.5245*** | .4617 | .6187 | **.7395*** | .6791 |

We use Fluent Speech Commands (FSC) [15] for training and in-domain evaluation. Each utterance has an action–object–location frame; FSC test evaluation contains 6,081 erroneous hypotheses from 2,777 utterance identifiers. We additionally use the Spoken Language Understanding Resource Package (SLURP) [16] to evaluate zero-shot transfer to a different dataset and task structure. SLURP uses a broader semantic representation containing scenario, action, and entity information. The strict SLURP evaluation contains 29,701 erroneous hypotheses from 1,573 identifiers. We further evaluate zero-shot transfer on SLUE-VoxPopuli [17], which contains natural parliamentary speech and named-entity annotations.

### 4.2. Supervision and Evaluation Targets

For evaluator-based downstream tasks, transcript-level failure follows Eq. (1), where task predictions are compared between the reference and hypothesis. Pairwise semantic criteria such as NLI are defined separately below.

**FSC supervision target.** For EC-WER supervision, we instantiate the task evaluator in Eq. (3) as $g_F$, consisting of three TF–IDF logistic-regression classifiers that predict the FSC action, object, and location. Each classifier achieves perfect accuracy on the 3,118 clean FSC validation references, yielding an exact full-frame accuracy of 1.0. We define the corresponding transcript-level failure target as $z_F := z_{g_F}$ using Eq. (1). Because $g_F$ also generates the edit-consequence labels used to train EC-WER, $z_F$ measures fidelity to the supervision evaluator rather than independent transfer.

**Independent FSC targets.** We evaluate transfer within FSC using two targets that do not contribute to EC-WER supervision or model selection. First, $g_B$ is an independently trained RoBERTa-base [18] classifier over 31 complete FSC task frames. It predicts every clean FSC test reference correctly but agrees with $g_F$ on only 76.5% of erroneous-hypothesis failure labels (Cohen's $\kappa = 0.476$). We define its failure target as $z_B := z_{g_B}$ using Eq. (1). Second, we use a pretrained RoBERTa-base NLI cross-encoder[1] and mark a reference–hypothesis pair as inconsistent unless entailment holds in both directions.

**Zero-shot SLURP targets.** We next apply the frozen FSC-trained metric directly to SLURP without SLURP-specific training or tuning. We evaluate three targets: full-frame failure under a frozen scenario–action–entity evaluator, intent failure under a separate RoBERTa-base 60-class intent classifier, and bidirectional NLI inconsistency. The evaluator-based full-frame and intent targets follow Eq. (1), while NLI uses the same pairwise criterion as in FSC. Neither EC-WER nor Direct Estimate uses SLURP data, evaluators, or targets for training or parameter selection.

**Zero-shot VoxPopuli target.** Finally, we evaluate named-entity recognition on VoxPopuli. Let $g_{\mathrm{NER}}$ denote the independently trained frozen NER evaluator and $Y_R^{\mathrm{gold}}$ the gold entity set for reference $R$. We retain only references for which $g_{\mathrm{NER}}(R) = Y_R^{\mathrm{gold}}$, preventing clean-reference evaluator errors from being attributed to ASR. On this retained set, we define $z_{\mathrm{NER}} := z_{g_{\mathrm{NER}}}$ using Eq. (1).

### 4.3. Training

We instantiate $P_\phi$ and $P_\theta$ as separate DeBERTa-v3-small sequence classifiers [19] with the same backbone architecture but different inputs and supervision targets. All supervision is generated from the FSC training split. We retain commands for which $g_F$ correctly predicts the annotated action–object–location frame and divide the 248 normalized command types into 198 training and 50 validation commands with no overlap. Synthetic noisy hypotheses are generated by applying one to three randomly sampled substitutions, deletions, or insertions to each command, with substituted and inserted words sampled from the training-command vocabulary.

For EC-WER, we recompute the WER alignment and evaluate Eq. (3) for each resulting edit, yielding 28,864 training and 3,626 validation edits. For Direct Estimate, each noisy reference–hypothesis pair is labeled with $z_F$, yielding 15,840 training and 2,000 validation pairs. Both students are trained for four epochs with batch size 16, learning rate $2 \times 10^{-5}$, and maximum sequence length 128.

### 4.4. Comparison Methods

WER serves as the nonsemantic baseline. We additionally compare against BERTScore distance [6], SemDist-style cosine distance [3], and Aligned Semantic Distance (ASD) [7]. For ASD, we use an English adaptation with `bert-base-uncased` [20] contextual token representations. We also evaluate the continuous SimCSE semantic-distance signal used for SASRED weak labeling [4], using the authors' `princeton-nlp/sup-simcse-roberta-large` encoder [21] and one minus cosine similarity between reference and hypothesis embeddings. We refer to this baseline as SimCSE (SASRED); it does not reproduce the full SASRED classifier, which additionally uses task-specific weak and human supervision. We also compare against $S^2$ER [9], an LLM-based metric that uses three-round bidirectional semantic-equivalence judgments. We use a Qwen3-32B-AWQ [22] judge with thinking disabled.

### 4.5. Metrics

We evaluate how well each ASR metric ranks hypotheses according to downstream failure. The positive class is a hypothesis that fails the corresponding downstream target. We report area under the precision–recall curve (AUPRC) and area under the receiver operating characteristic curve (AUROC), with higher values indicating better ranking performance. Because downstream failures are imbalanced, we treat AUPRC as the primary metric and AUROC as a complementary measure. Because $S^2$ER produces a binary semantic-error decision rather than a continuous ranking score, its AUPRC and AUROC are computed directly from this binary output. Statistical comparisons use 5,000 paired cluster-bootstrap replicates with two-sided tests at $p < 0.05$.

[1] `cross-encoder/nli-roberta-base`

**Table 2**: Edit-level consequence prediction. SLURP is evaluated zero-shot using the FSC-trained EC-WER student.

| Data | Edits | AUPRC | AUROC |
|---|---|---|---|
| FSC synthetic | 3,626 | .858 | .925 |
| SLURP real ASR | 105,266 | .876 | .728 |

**Table 3**: Context-input ablation for edit-consequence prediction. The context-sensitive subset contains 499 validation edits where the same focal edit receives different labels across noisy contexts.

| | All edits | | Context-sensitive | |
|---|---|---|---|---|
| Input | AUPRC | AUROC | AUPRC | AUROC |
| Edit only | .7174 | .8084 | .5807 | .6462 |
| Reference + edit | .7459 | .8443 | .5857 | .6564 |
| Hypothesis + edit | **.8608** | .9184 | **.7962** | .8454 |
| Full context | .8576 | **.9250** | .7937 | **.8567** |

## 5. RESULTS

### 5.1. Transcript- vs Edit-Level Distillation

Table 1 compares transcript- and edit-level consequence distillation. Direct Estimate achieves higher AUPRC on 4 of 7 downstream targets, while EC-WER achieves higher AUPRC on the remaining. Paired cluster-bootstrap tests show that most of these differences are statistically significant, while the small EC-WER advantage on SLURP full-frame failure is not.

This tradeoff yields a practical selection rule: Direct Estimate should be preferred when the only goal is ranking hypotheses by downstream-failure risk, while EC-WER is preferable whenever the application also requires knowing which edits drove that risk, such as for ASR model debugging or analysis.

### 5.2. Zero-shot Transfer

Both consequence-distillation models are trained only on FSC yet transfer without retraining to multiple SLURP targets and VoxPopuli NER. On SLURP, EC-WER achieves higher AUPRC on full-frame and intent failure, while Direct Estimate performs better on NLI; Direct Estimate also performs better on VoxPopuli NER.

To directly test whether EC-WER's edit-level attribution transfers beyond FSC, we counterfactually repair errors in real SLURP ASR hypotheses and measure whether each repair changes the frozen SLURP full-frame prediction. Table 2 shows that EC-WER's edit-level consequence scores transfer zero-shot to SLURP.

### 5.3. Ablations

Among consequential edits, 71.6% of repairs recover the correct frame and only 4.7% turn a correct prediction into an incorrect one, indicating that prediction changes usually reflect beneficial corrections.

Table 3 shows that edit consequence is not a fixed property of the edited words. The context-sensitive subset contains 499 edits for which the same focal edit receives different consequence labels across noisy hypotheses. Conditioning on the noisy hypothesis substantially improves prediction over using the edit alone, showing that consequence depends on the surrounding recognition errors.

**Table 4**: Comparison with existing ASR evaluation metrics. FSC uses the supervision-aligned $g_F$ target, while SLURP uses zero-shot full-frame evaluation. $^*$ indicates a statistically significant improvement over the strongest competing semantic baseline ($p < 0.05$).

| Data | Metric | AUPRC | AUROC |
|---|---|---|---|
| FSC | WER [1] | .6093 | .7111 |
| | SemDist [3] | .5136 | .6266 |
| | BERTScore [6] | .5828 | .6947 |
| | ASD [7] | .5164 | .6417 |
| | SimCSE (SASRED) [4] | .6132 | .7407 |
| | $S^2$ER [9] | .4287 | .5987 |
| | Direct Estimate | .7996 | .8319 |
| | EC-WER | **.8245**$^*$ | **.9188**$^*$ |
| SLURP | WER [1] | .9486 | .8126 |
| | SemDist [3] | .9466 | .7929 |
| | BERTScore [6] | .9555 | .8234 |
| | ASD [7] | .9534 | .8260 |
| | SimCSE (SASRED) [4] | .9619 | .8416 |
| | $S^2$ER [9] | .8535 | .5846 |
| | Direct Estimate | .9640 | .8549 |
| | EC-WER | **.9644**$^*$ | **.8554**$^*$ |

Full context achieves the highest AUROC, while hypothesis-plus-edit achieves slightly higher AUPRC, indicating that the noisy hypothesis supplies the dominant contextual signal.

We also compared random corruption with ASR-generated supervision. A size-matched random-corruption model outperformed a model trained on pooled errors from three independent ASR systems on five of seven targets, while the ASR-generated model performed better on the supervision-aligned $g_F$ target. This suggests that random corruption transfers more broadly across evaluators and tasks.

### 5.4. Interpretation and Semantic Baselines

The results show complementary behavior between the two granularities. Direct Estimate provides strong transcript-level failure ranking, while EC-WER retains competitive performance and additionally attributes consequence to individual WER edits. The context ablations further show that edit consequence depends on the surrounding recognition errors rather than only on edit identity.

Table 4 compares consequence distillation with existing ASR evaluation metrics. Because Direct Estimate and EC-WER use FSC consequence supervision whereas the semantic baselines do not, we treat FSC as a fidelity test. However, SLURP provides a zero-shot comparison, where EC-WER outperforms the semantic baselines on strict full-frame failure while retaining edit-level attribution.

## 6. CONCLUSION

We presented consequence distillation for ASR evaluation at transcript and edit granularities. Direct Estimate provides a strong predictor of downstream task failure, while EC-WER uses counterfactual supervision to attribute consequence to individual substitutions, deletions, and insertions. Across independent FSC targets and zero-shot transfer to SLURP and VoxPopuli, the results reveal a granularity tradeoff, offering practitioners a clear choice: Direct Estimate for pure downstream-failure prediction, and EC-WER for interpretable, edit-level error diagnosis.

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## 9. COMPLIANCE WITH ETHICAL STANDARDS

This research used only publicly available datasets and involved no new data collection or experiments with human participants. No new personally identifiable information was collected, and no additional ethical approval was required.